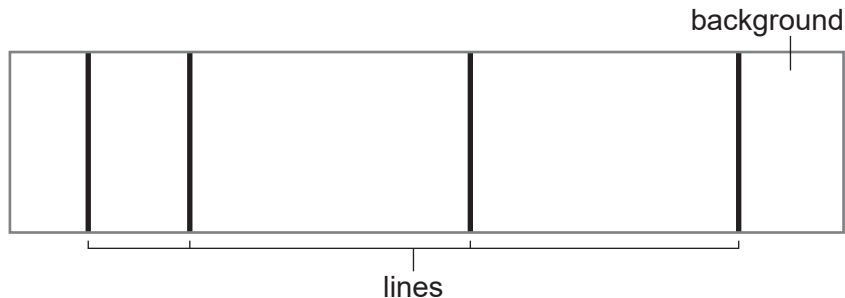


## E.1 / PAPER 1 / QUESTION 8

28 April 2022 / TZ2 / HL Paper 1 / original Q22 / 2222-6513

8. White light is emitted from a hot filament. The light passes through hydrogen gas at low pressure and then through a diffraction grating onto a screen. A pattern of lines against a background appears on the screen.



What is the appearance of the lines and background on the screen?

|    | Lines | Background |
|----|-------|------------|
| A. | dark  | black      |
| B. | white | coloured   |
| C. | white | black      |
| D. | dark  | coloured   |